



**TEMASEK JUNIOR COLLEGE
PRELIMINARY EXAMINATION
JC 2 2018**

CANDIDATE
NAME

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CENTRE
NUMBER

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INDEX
NUMBER

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CLASS

C	G			/	1	7
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H2 BIOLOGY

9744/03

Paper 3 Long Structured and Free-response Questions

**Thursday 13 September 2018
2 hours**

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Section A

Write your name, Centre Number, index number and class in the spaces at the top of the page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graph.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the separate Writing Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Q1	/14
Q2	/13
Q3	/23
Q4 / 5	/25
Total	/75

This document consists of **21** printed pages and **1** blank page.

- 1 Scientists investigated three genes, **C**, **D** and **E**, involved in controlling cell division. They studied the effect of mutations in these genes on the risk of developing lung cancer.

The scientists analysed genes **C**, **D** and **E** from healthy people and people with lung cancer.

- If a person had a normal allele for a gene, they used the symbol **N**.
- If a person had two mutant alleles for a gene, they used the symbol **M**.

They used their data to calculate the risk of developing lung cancer for people with different combinations of **N** and **M** alleles of the genes. A risk value of 1.00 indicates no increased risk.

Table 1.1 shows the scientists' results.

Table 1.1

Gene C	Gene D	Gene E	Risk of developing lung cancer
N	N	N	1.00
M	N	N	1.30
N	N	M	1.78
N	M	N	1.45

N = at least one copy of the normal allele is present

M = two copies of the mutant allele are present

- (a) Suggest the relative importance of the mutant alleles of genes **C**, **D** and **E** on the risk of developing lung cancer. Explain your answer. [3]
1. Two copies of mutant allele E produces the highest risk of 1.78 [1]
 2. While two copies of mutant allele D produces the second highest risk / second lowest of 1.45 [1]
 3. Two copies of mutant allele C produces the lowest risk of 1.30 [1]

Chemotherapy is the use of a drugs, such as vinblastine, to treat cancer. The drug kills dividing cells. Fig. 1.1 shows the number of healthy cells and cancer cells in the blood of a patient receiving chemotherapy. The arrows labelled F to I show when the drug was given to the patient.

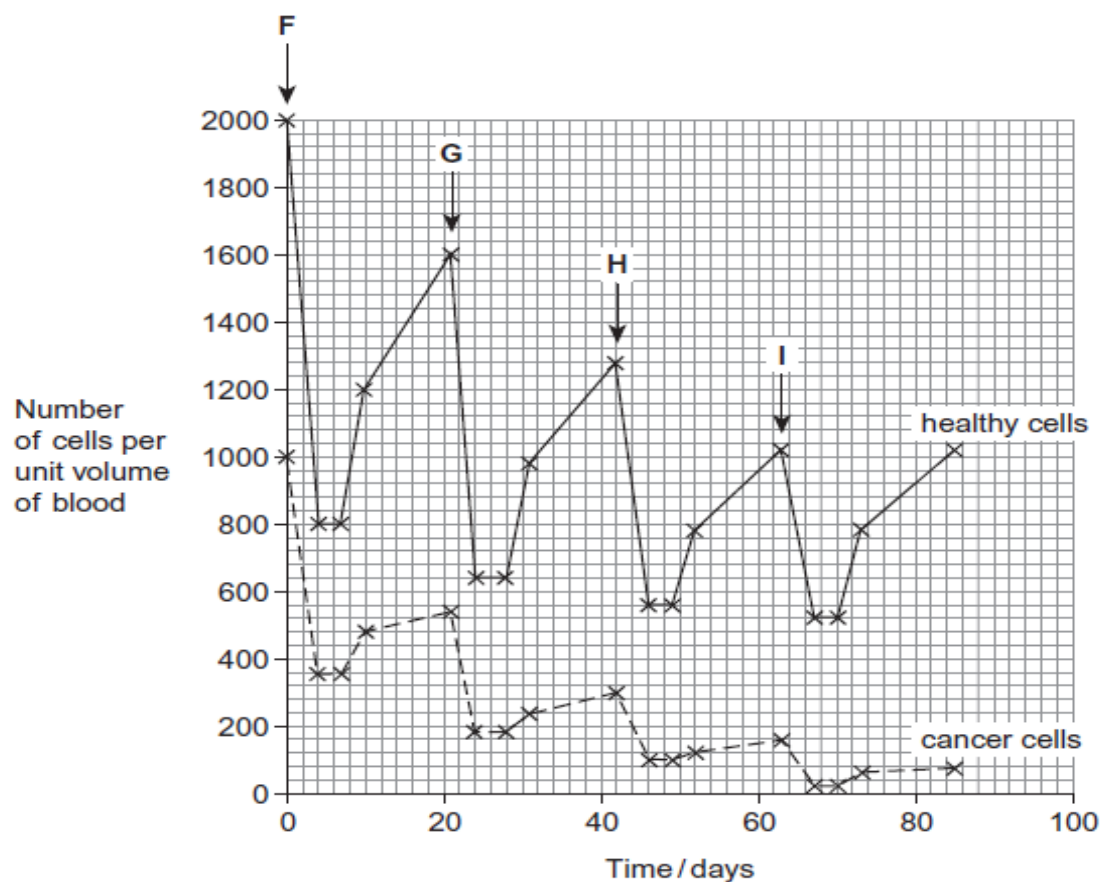


Fig. 1.1

- (b) Calculate the rate at which healthy cells were killed between days 42 and 46.

$$1280 - 560 = 720$$

$$720 \div 4 = 180$$

180 cells killed per unit volume of blood per day

[1]

- (c) Describe **two** similarities and **one** difference in the response of healthy cells and cancer cells to the drug between times **F** and **G**. [3]

Similarities:

Any two

1. Both show a steep decrease in number of cells per unit volume from day 0 to 4.
 2. For both, number of cells per unit volume remained constant from day 4 to 7.
 3. Both show a steep increase in number of cells per unit volume from day 7 to 21.
 4. Both show and overall net decrease [QF] in the number of cells per unit volume at time point G
 5. Both have similar pattern of decrease, remain the same and then increase
- OR

Both showed a fluctuation of decrease, remaining the same and then increasing

Differences:

6. There is greater decrease in number of healthy cells / more healthy cells were killed than cancer cells after the drug is given [QF].
7. There is greater increase in number of healthy cells during recovery period / more healthy cells are replaced during recovery period than cancer cells [QF].

- (d) More cancer cells could be destroyed if the drug was given more frequently. Suggest why the drug was **not** given more frequently. [2]

1. Too many healthy cells are killed after each dose of drugs
- OR
- It takes time to replace the number of healthy cells
2. The person may die
- OR
- have severe side effects if the drug was given more frequently / become immunocompromised

- (e) State **two** ways in which cancer cells differ from normal healthy cells. [2]

Point of comparison	Cancer cell	Normal healthy cell
1. Apoptosis	Does not experience apoptosis	Undergoes apoptosis
2. Cell division	Uncontrolled	Controlled with various checkpoints.
3. Contact inhibition	Not affected	Affected.
4. Specialized cell	do not become specialized, but remain immature.	Able to differentiate into specialized cells.
5. Able to stimulate angiogenesis	yes	no
6. Able to metastasize	yes	no
7. Telomerase gene	expressed	Not expressed

Vinblastine disrupts the formation of the spindle apparatus during mitosis.

- (f) Explain how vinblastine exerts its effect as an anti-cancer drug. [3]

1. Cancer cells carry out mitosis repeatedly / uncontrolled cell division

During mitosis (role of spindle fibres):

2. Spindle fibres cannot attach to the kinetochore of the centromere of the chromosome.
3. Spindle fibres cannot align chromosomes at the equator of cell during metaphase
4. Spindle fibres cannot separate sister chromatids at the equator during anaphase
5. The disruption of spindle fibres formation means that metaphase and anaphase cannot take place.
6. Mitosis stops and the cancer cells stop dividing, some cells undergo apoptosis.

[Total: 14]

- 2 The area over which the Arctic ice sheet extends varies throughout the year. Fig. 2.1 shows the variation in the extent of the Arctic ice sheet for the months of July to November for the years 1979 and 2009.

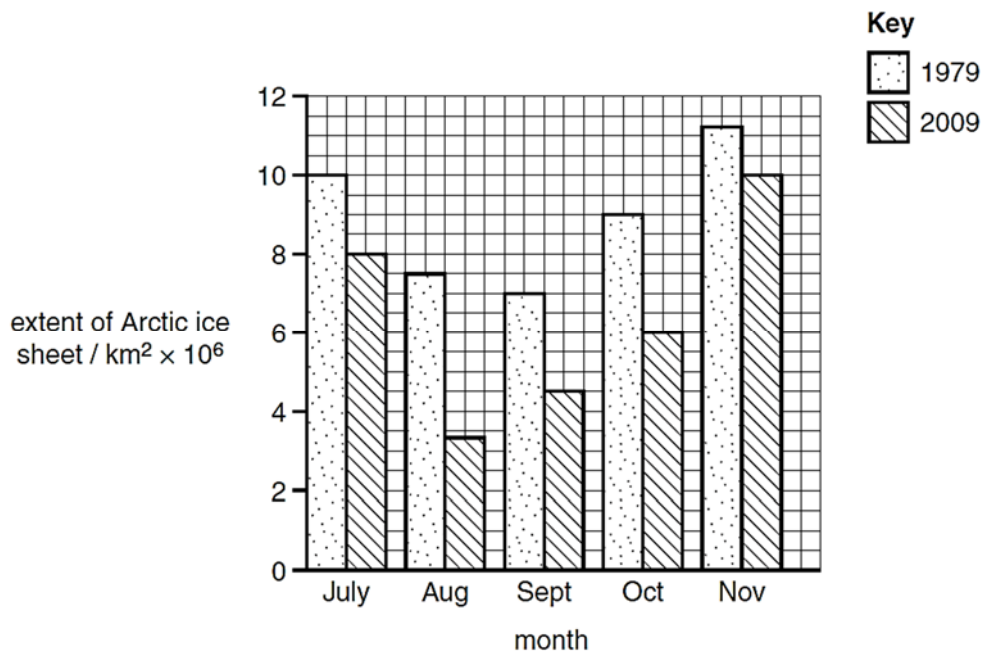


Fig. 2.1

- (a) Calculate the percentage reduction in the area over which the ice sheet extends between 1979 and 2009 for the month of September. [1]

$$\text{Percentage reduction} = \frac{(4.5 \times 10^6 - 7 \times 10^6)}{7} \times 100\% = 35.7\%$$

- (b) Suggest reasons for the reduction in the Arctic ice sheets from 1979 to 2009. [2]

Climate Change Part I Lecture Notes p.17

1. Increased rate of deforestation / clearing of forest / reduction of carbon sink [1]
2. Increased burning of fossil fuels / energy consumption for homes, industries or transport [1]
3. Increased rearing of livestock [1]
4. Increase in concentration of greenhouses gases (carbon dioxide, methane) in the atmosphere results in more heat trapped, hence resulting in an increase in air, land and/ or sea temperature, leading to the melting of Arctic ice sheets / decrease in snowfall / ice sheets forms slower. [1]

- (c) The polar bear, *Ursus maritimus*, moves across the Arctic ice sheet to hunt prey such as seals. When seals surface to breathe at cone-shaped breathing holes on the sea ice, a hunting polar bear which is waiting by the breathing hole will smack the head of the seal with both of its front paws to stun it, before biting and dragging the seal onto the ice. This method of still-hunting minimizes energy consumption and is the most successful strategy of hunting.

In 2008 the government of the USA classified *U. maritimus* as an endangered species because it is under the threat of extinction.

Suggest how climate change could have caused *U. maritimus* to become an endangered species. [2]

1. Global warming in the Arctic resulted in melting sea ice
 2. and reduction in extent of ice sheets / loss of habitat for both seals and polar bears.
 3. It also leads to more [i.e. initial stage] / less or bigger breathing holes [i.e. later stages when the ice melts and the breathing holes fuse]
- OR
- The seals migrate to other cooler places / no longer breathe at breathing holes,
4. therefore the polar bears are unable to still-hunt seals (only source of food) / use other hunting strategies that increases energy consumption, hence they starve and die.

Climate change also affects plants.

Plants can be categorized based on the way they photosynthesize. Most plants are C₃ plants because their first photosynthetic product is a three carbon compound. Examples of C₃ plants include barley, oats, potato, rice, and wheat commonly grown in temperate regions.

On the other hand, C₄ plants produce a four-carbon compound as their first photosynthetic product. Examples of C₄ plants are common grass crops of tropical regions, such as maize, millet, sorghum and sugarcane.

The rate of carbon dioxide uptake at a range of carbon dioxide concentrations by barley, a C₃ plant, and sugar cane, a C₄ plant, were compared at two temperatures.

The results of the experiment are presented in Fig. 2.2.

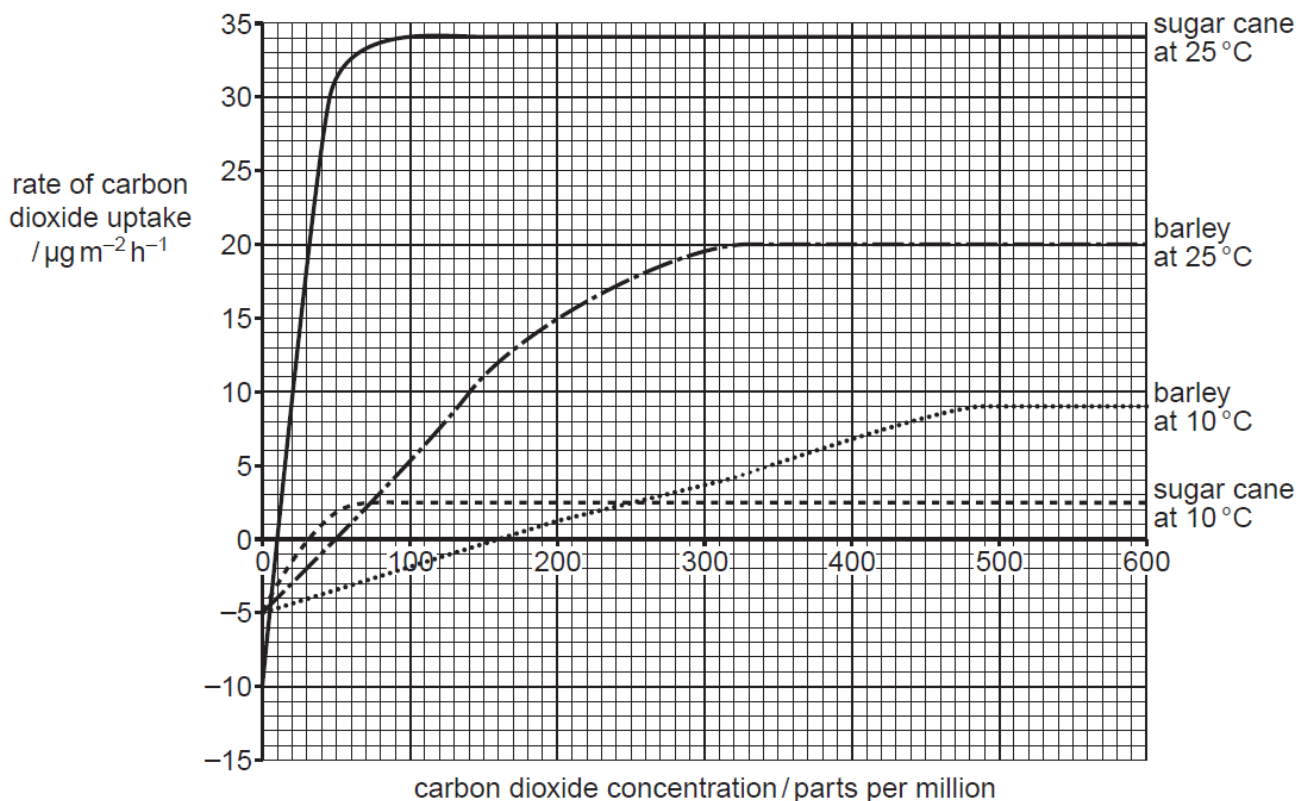


Fig. 2.2

The current carbon dioxide concentration in the atmosphere is more than 400 parts per million and it is likely to increase in the future. It is widely believed that the carbon dioxide concentration of the atmosphere affects the global mean surface temperature which in turn changes rainfall patterns.

Table 2.1 shows other data regarding three C3 and three C4 plants which are important crops.

Table 2.1

crop	mass of water absorbed per gram dry mass produced/ g
rice	682
potato	575
wheat	542
maize	350
millet	285
sorghum	204

With reference to Fig. 2.2 and Table 2.1,

- (d) discuss the likely impact of the predicted changes in carbon dioxide concentration, global temperatures and rainfall patterns on the global distribution of C3 and C4 plants. [4]

1. Increased in carbon dioxide concentration will likely increase temperature.
2. At high carbon dioxide concentration of 500 parts per million and high temperature of 25°C, the rate of photosynthesis for C4 plants / sugar cane at 34 $\mu\text{gm}^{-2}\text{h}^{-1}$ is HIGHER than that of C3 plants/ barley at 20 $\mu\text{gm}^{-2}\text{h}^{-1}$. [1]
 [Accept: Maximum rate of photosynthesis. Reject: Peak]
 [Accept: Any high value of carbon dioxide concentration more than 400 ppm]
3. Hence, both plants will grow well, but C4 plants are better adapted than C3 plants in hotter areas and their population will likely increase/ OWTTE.
 [Accept: Reference to latitude (tropical / temperate) or altitude]
4. Increased temperatures may result in lower rainfall in some places.
 [Accept: Higher rainfall]
5. C4 plants absorb between 204 to 350g of water which is LESS than C3 plants between 542 and 682g. [1]
6. Hence, C4 plants are better adapted than C3 plants in drier areas and their population will likely increase / OWTTE.
 [Accept: Higher rainfall, C3 plants will grow better OR both plants will grow well]
7. [Additional] However, predicted change in temperature over the next century is only small, therefore it may not make a lot of difference.

Climate change potentially affects the spread of diseases. Fig. 2.3 shows the worldwide distribution of dengue.

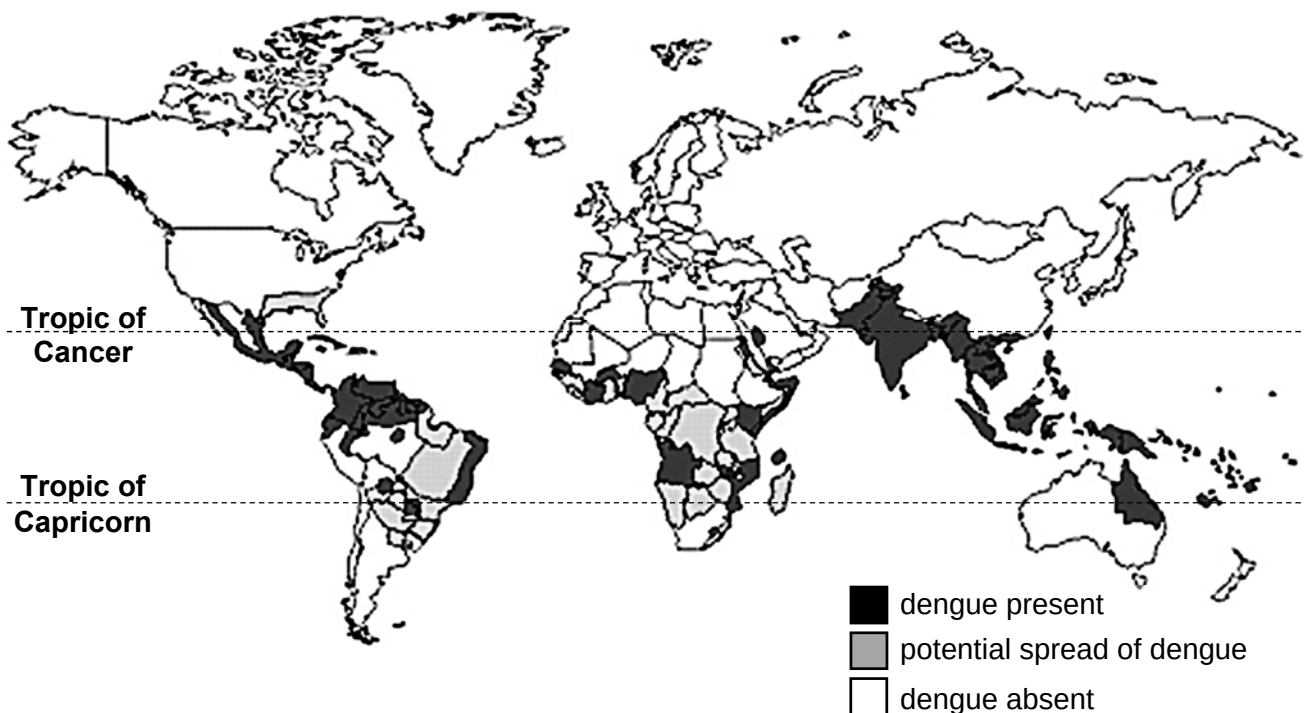


Fig. 2.3

(e) Describe how dengue is transmitted. [2]

1. FEMALE
2. Aedes mosquito
[Reject: virus or bacteria]
3. takes a blood meal from an infected person, and bites/ feeds on an uninfected person/ OWTTE. [1]
4. Dengue virus is transmitted in mosquito's saliva.

Unlike dengue, influenza is found across the whole world.

(f) Explain why dengue shows the distribution pattern shown in Fig. 2.3, but influenza is found everywhere. [2]

1. Dengue is vector-borne disease (i.e. caused by dengue virus and transmits / reproduces within Aedes mosquito)
2. which lives within the Tropics of Cancer and Capricorn [1]
OR hot and humid areas (at temperatures above 20°C) at a favourable temperature range for breeding / reproduction of mosquitoes,
3. whereas influenza is air-borne disease / transmitted by respiratory droplets/ coughing/ sneezing,
4. and it is not limited by the range of the vector (i.e. hot and humid conditions) to be transmitted / spread to other parts of the world by infected travelers.
5. [Additional] Mosquitoes are prevalent in tropical region due to poor / non-existent mosquito control programmes/ OWTTE.
6. [Additional] Mosquitoes may also be resistant to insecticides.

[Total: 13]

3 Hepatitis is the inflammation of the liver and can be caused by a number of different hepatitis viruses. Presently, the only effective vaccines available are for hepatitis A and B.

(a) Outline the immune response that leads to the production of antibodies after vaccination. [3]

1. Vaccination trigger active immunity / active immune response
2. Vaccine contains hepatitis antigen / hepatitis pathogen / virus;
3. Antigen presenting cells / macrophage present antigen to specific naive CD4 T cells and naive B cells
4. Helper T cells secrete cytokines
Many students did not include this point.
5. Specific B cells activated to proliferate and differentiate into plasma cells and memory B cells.
6. Plasma B cells produce antibodies

(b) Describe how plasma cells produce and release antibodies. [5]

1. Transcription of light and heavy chain gene / antibody gene produces mRNA
2. Pre-mRNA processing take place.
3. Ribosomes on rER translate mRNA
4. Anticodon of tRNA base pair with codons on mRNA and bring amino acids / ensure amino acids join in correct sequence.
5. Heavy and light chains of antibodies move into rER lumen.
[to award if student mention heavy and light chains later]
6. These are enclosed in ER/ transport vesicle which pinch / bud off from ER
7. to cis face of Golgi apparatus (GA).
8. GA chemically modifies, sorts and transports antibodies.
9. Heavy and light chains are joined by disulfide bonds and glycosylated as they move through the GA
10. The secretory vesicle buds off from trans-face of GA, travels along microtubules of cytoskeleton,
11. fuse with the plasma membrane. Thus, releasing the antibodies out of the cell via exocytosis.

Scientists observed that liver cells damaged by hepatitis infection switch on a gene known as the *Fas* gene, which caused infected liver cells to self-destruct.

This finding has led to pioneering research which produced a successful treatment for hepatitis in mice. The *Fas* gene was silenced using the technique of RNA interference.

This involved injecting mice infected with hepatitis with RNA molecules of 21 to 23 nucleotides in length. The sequence of this small interfering RNA (siRNA) matched part of the *Fas* gene. Once in the liver cell the two strands of the siRNA are separated so that one strand binds to the mRNA transcript of the *Fas* gene.

This caused the mRNA to be destroyed by enzymes, therefore preventing the gene product from being made. This therapy prevented liver cell death and considerably increased the survival of mice with hepatitis.

(c) (i) Describe **one** way in which the function of mRNA differs from that of DNA. [1]

1. mRNA is translated / used to synthesize protein while DNA is transcribed / used to synthesize mRNA;
OR
mRNA is used to synthesize protein while DNA is for the storage of genetic information

OR

2. mRNA contain short-term genetic information while DNA contain long term genetic information

(ii) Suggest **one** way in which the structure of siRNA differs from mRNA. [1]

1. siRNA has fewer nucleotides than mRNA / only matches part of gene.

OR

2. siRNA double-stranded while mRNA is single-stranded

(d) Describe how one strand of the siRNA can bind to the mRNA of the *Fas* gene. [2]

1. Via complementary base-pairing between purines and pyrimidines; [1]
2. Adenine with uracil with 2 hydrogen bonds and;
3. cytosine with guanine with 3 hydrogen bonds

The technique of RNA interference has also been used to slow down replication of HIV (Human Immunodeficiency Virus) *in vitro*. This is an important breakthrough in the treatment of AIDS as many countries are hit by the epidemic.

The siRNA is attached to a carrier molecule which binds to HIV protein on the plasma membrane of infected cell. This allows carrier with siRNA to enter human cell.

siRNA sequences that match the RNA genome of HIV can be used to trigger destruction of this RNA, preventing HIV from multiplying.

(e) The siRNA would **only** affect gene expression in cells infected with HIV. Suggest **one** reason why. [1]

1. Only infected cells have HIV protein on surface;
 2. So carrier only attaches to/specific to these cells/siRNA can only enter these cells
- OR
3. Only infected cells contain RNA of HIV
 4. Base sequence of siRNA is only complementary to the HIV RNA to destroy it.

Another approach is to use RNA interference to silence genes for cell surface receptors, such as the CD4 and CCR5 molecules on human white blood cells.

If these genes are not expressed, HIV cannot bind to and infect the white blood cells. Table 3.1 summarizes some information regarding the two cell surface receptors used by HIV to bind to and infect white blood cells.

Table 3.1

	cell surface receptor	
	CD4	CCR5
Type of cell with this receptor	T lymphocyte white blood cells which divide by mitosis	Macrophage cells which are long-lived and do not undergo mitosis

Experiments have been carried out where,

- siRNAs matching the CD4 mRNA were introduced into test tube populations of T lymphocytes;

- siRNAs matching the CCR5 mRNA were introduced into test tube populations of macrophages.

In both cases HIV was present and the presence of the siRNAs reduced its replication.

- (f) Using Table 3.1, suggest with reasons which of the two test tube experiments would have a greater reduction in HIV replication. [2]

Test tube containing CCR5 / macrophages

1. only one treatment needed for macrophages / CCR5;
2. because siRNAs has longer effects in long-lived cells;
3. whereas siRNAs diluted / fewer per cell when lymphocytes divide;
4. repeat treatments needed for lymphocytes / CD4;

Antibiotics are prescribed to people who have HIV/AIDS for the treatment of secondary infections such as bacterial infections.

- (g) Describe the mode of action of antibiotics, such as penicillin, on bacteria. [3]

1. Penicillin binds irreversibly / inhibits to transpeptidase
2. thus inhibiting the cross-linking of two peptidoglycan chains.
3. Penicillin also stimulates the release of autolysins
4. and make small pores in the existing cell wall.
5. The cell wall of dividing bacterium weakens
6. Osmotic lysis occurs
Reject : autolysis

- (h) Explain why antibiotics are prescribed to treat secondary infections, but not HIV infection. [2]

1. People with HIV are very susceptible to bacterial infections due to weakened immune system
2. Antibiotics are only effective against bacteria
Reject: microbes or micro-organisms instead of stating bacteria
OR
3. Antibiotics not effective against viruses,
Reject: this is only accepted if point four is present
Reject: antibiotics prevent infection
Reject: if did answer did not include "virus"
4. Viruses do not have cell walls, ribosomes or cell membranes that antibiotic work on
Note: must state specific organelles.
Reject: vague mention of cell machinery or virus is non-cellular.
OR
5. viruses are within cells, idea that antibiotics cannot reach them.

Antibiotic resistance could develop and the genes for antibiotic resistance could be transmitted between bacteria. Table 3.2 shows features of gene transmission.

Table 3.2

Statement	Vertical	Horizontal
Gene is replicated	✓	✓
Gene can be passed to other species of bacteria		✓
Involves conjugation		✓

- (i) Complete Table 3.2 by putting a tick in the box if the statement is correct for vertical or horizontal gene transmission. [1]

Apart from the devastating effects of HIV, in 2014, parts of West Africa were hit by an epidemic of Ebola fever. Most people who caught the disease died.

Scientist attempted to genetically synthesize an antibiotic as a possible drug to target the Ebola glycoprotein.

This drug was **only** used to treat two Americans who had been working as medics in Africa. Its use was controversial because the drug had not been tested on humans. At the time there were only a few doses of the drug available.

- (j) (i) Suggest a reason why the decision was made to use the drug, even though it had not been tested. [1]
- The Ebola infected individual would have died anyway;
- (ii) Apart from the fact that drug had not been fully tested, give **one** reason why using the drug in the way described could be considered as unethical. [1]
- The Americans had no more right to treatment than the Africans/owtte;

[Total: 23]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must set out in sections **(a)** and **(b)**, as indicated in the question.

EITHER

- (a)** Describe the causes of variation and its biological importance and how variations are preserved in a population. [10]

CAUSES OF VARIATION		
A. Effects of Mutation [MAX : 2 marks]		Remarks
A1	▪ <u>Gene mutation</u> is <u>change</u> in <u>sequence</u> of <u>nucleotides</u> ▪ E.g. addition / substitution / deletion	▪ Reject: change in sequence of genes
A2	<u>Give rise</u> to <u>new alleles</u> [ONLY WAY]	▪ Note: Evolution does NOT give rise to new alleles. ▪ Reject: new genes are formed. ▪ Do NOT confuse genes and alleles
A3	<u>Effect</u> on <u>dominant allele</u> verses recessive alleles → giving rise to <u>different genotypes</u> and <u>phenotypes</u>	
A4	<u>Effect</u> of <u>mutation on polypeptide</u> / protein → change shape / change function	▪ Reject focus on non-functional protein
A5	<u>Chromosome mutation</u> - <u>change</u> in <u>number</u> and <u>structure</u> of <u>chromosomes</u>	
A6	<u>Multiple alleles</u> → many <u>different genotypes</u> possible	
B. Effects of meiosis [MAX : 2 marks]		
B1	<u>Crossing over</u> between <u>homologous chromosomes</u> during <u>prophase I</u>	▪ Many students wrote "crossing over between non-sister chromatids" ▪
B2	<u>Independent arrangement</u> and <u>separation</u> of <u>homologous chromosomes</u> at <u>metaphase I</u> and <u>anaphase I</u> respectively	▪ Reject: if left out "respectively"
B3	<u>Independent arrangement</u> of <u>chromosomes</u> at <u>metaphase II</u> and <u>separation of chromatids</u> at <u>anaphase II</u>	▪ Reject if students combine the metaphase and anaphase events – referring only to chromatids or chromosomes ▪ Reject if students combine metaphase I and metaphase II; then anaphase I and anaphase II
B4	<u>Random fusion</u> of <u>gametes</u>	▪ Do NOT confuse random fusion of gametes with random mating.
B5	<u>Give rise</u> to <u>new combinations of alleles</u>	▪ This must be written at least once.
C. Effects of Environment [MAX : 1 mark]		
C1	<u>Affects traits</u> showing <u>continuous variation</u>	
C2	<u>Presence</u> or <u>absence</u> of <u>nutrients</u> – affect animals	

C3	<u>Presence</u> of <u>Disease</u> – affects both plants and animals	
C4	<u>Light</u> – affects <u>plants</u> [amount of chlorophyll, thickness of leaves] <u>More light</u> – <u>more chlorophyll</u> needed to absorb UV rays More light – <u>leaf</u> can be <u>thicker</u> . Spongy mesophyll cells can still absorb sufficient light energy.	
C5	<u>Temperature</u> 5a. affects <u>animals</u> e.g. length of hair; colouration 5b. affects <u>plants</u> e.g. needle shape leaves to reduce water loss	
C6	Presence of <u>harmful chemicals</u> → cause mutations / diseases	
D. BIOLOGICAL IMPORTANCE OF VARIATION [MAX : 2 marks]		
D1	Variation <u>enables adaptation</u> – some are better adapted because of new alleles with new phenotypes	▪ Genetic variation does not ensure individuals are well adapted.
D2	Natural selection - <u>Variation within a population</u> is an <u>essential</u> pre-condition <u>for</u> evolution through <u>natural selection</u> .	▪ Selection pressure does not cause different species to have different alleles
D3	Natural Selection <u>selects existing favorable phenotype</u> / <u>selective advantage survive</u> → <u>reproduce</u> → <u>pass allele to offspring</u>	▪ Evolution does not give rise to new alleles / genes
D4	<u>Speciation</u> – when <u>geographical isolation</u> exists, new species formed over a <u>long period of time</u> with	
E. PRESERVING VARIATION [MAX : 3 marks]		
Relating to Diploidy:		
E1	<u>Diploid organisms</u> carries a <u>large amount of variation</u> in the form of <u>recessive alleles</u> in <u>heterozygotes</u> .	
E2	This <u>maintains genetic variation</u> in the form of <u>hidden recessive alleles</u> in <u>heterozygotes</u> / <u>Heterozygotes</u> maintain a <u>huge pool of alleles</u> that may not be suited for present conditions but could bring new benefits when the environment changes.	Awarded only if point 3 is written
E3	The <u>rarer</u> the <u>recessive allele</u> , the <u>greater</u> the <u>degree of protection</u> from natural selection /	
Relating to Heterozygote advantage:		
E4	Individuals who are <u>heterozygous</u> at a particular locus have <u>greater ability to survive</u> and <u>greater reproductive success than</u> any <u>homozygous</u> types.	Many students are not able to provide proper explanations
E5	This <u>maintains two or more alleles</u> at that locus by natural selection.	
E6	E.g. The <u>heterozygote</u> (HbA ^H HbS) is <u>better able to survive than</u> either of the two <u>homozygotes</u> in the <u>presence of malaria</u> .	
Relating to Frequency Dependent Selection:		
E7	<u>Frequency-dependent selection</u> - type of <u>balancing selection</u> that <u>maintains two different phenotypic forms</u> in a population.	

E8	The <u>selective advantage</u> of a <u>phenotype decreases</u> when it <u>becomes too common</u> .	
E9	Provide a brief description of Batesian mimicry or Predator-prey interactions	
Relating to Neutral Variation		
E10	Neutral variation – provides a source of variation which can be of selective advantage when environment change	
[MAX : 1 mark each]		
F.	GENETICALLY MODIFIED ORGANISMS - new genes leads to new combinations of phenotypes in genetically modified organism	▪ Evolution does not give rise to new alleles / genes
G.	FOUNDER'S EFFECT / BOTTLENECK EFFECT - allele frequency change due to chance event - alleles are loss	
H.	DIFFERENTIAL GENE EXPRESSION - Differential gene expression → changes phenotypes → some are better able to adapt to environment.	
	QWC : paragraphing + A2 + B5 + D2 + E	
	Total marks : A – 2 B – 2 C/F/G/H – 1 D – 2 E – 3	

(b) Justify the claim that all living organisms on Earth depend on phosphate.

[15]

Justify the claim that all living organisms on Earth depend on phosphate. [15]		
Big Idea: Membrane [max 3]		
A1	<u>Phospholipid</u> molecule made up of 1 glycerol, 1 <u>phosphate</u> and 2 fatty acids chains helps in the formation of cell membrane	
A2	The <u>hydrophobic core/ hydrophilic region</u> allows the <u>hydrophobic boundary</u> to <u>exist</u> in an <u>aqueous</u> environment. OR idea of compartmentalisation	
A3	<u>Control movement of substances</u> , exocytosis/endocytosis or maintenance of a <u>constant internal environment</u> within the cell/ organelle/ <u>maintenance of optimal/ high concentrations of reactants</u> at specific sites	
A4	The <u>fluidity</u> of the phospholipid bilayer allows the formation; (eg of structures) • Transient pores • Pseudopodia For transport of substances into the cell	
Big Idea: ATP [1 mark for different categories]		
B1	Phosphate found in ATP	
B2	Energy is stored in the <u>bonds</u> of <u>ATP</u> which are <u>broken/ hydrolyse/ currency of energy in the body</u>	

B3	<u>ATP</u> is hydrolysed to <u>actively pump substances</u> across the membrane during <u>active transport</u>	
B4	<u>ATP</u> is hydrolysed to provide energy for <u>amino acid activation</u>	
B5	<u>ATP</u> is hydrolysed to provide energy for <u>post translational modification of amino acids</u>	
B6	<u>ATP</u> used for phosphorylation of kinases/enzymes	
B7	<u>ATP</u> can be converted to cAMP by adenylyl cyclase	
Big Idea: GTP [max 3]		
C1	Phosphate found in GTP	
C2	GTP provides energy for the translocation process of translation	
C3	GTP serves as intermediate for formation of ATP during substrate level phosphorylation	
C4	GTP present in GPCR, for activation of GPCR	
Big Idea: Photosynthesis [max 3]		
D1	Guiding principle: name of molecule + structure + role <u>Ribulose-1,5-bisphosphate</u> is required for <u>carbon fixation</u> to form <u>glycerate-3-phosphate</u> .	
D2	<u>NADP⁺</u> is the <u>final electron acceptor</u> of <u>photophosphorylation</u> <u>OR</u> is essential to <u>reduce glycerate-3-phosphate</u> to <u>glyceraldehyde-3-phosphate</u> in <u>Calvin cycle</u> .	
D3	<u>Glyceraldehyde-3-phosphate</u> is required to synthesize <u>glucose</u> , <u>starch</u> , and <u>cellulose</u>	
Big Idea: Nucleotides [no max marking]		
E1	Nucleotide – Phosphate group, pentose sugar, nitrogenous base	
E2	Phosphate group required for formation of phosphodiester bonds / sugar-phosphate backbone	
E3	Nucleic acids (e.g. <u>DNA</u>) <u>stores genetic information</u>	
E4	and are <u>inherited</u> by the <u>offspring</u> to ensure the <u>continuity of the species</u>	
E5	Phosphate groups negatively charged, allowing for wrapping of DNA around histones (stability of DNA)	
E6	mRNA codes for protein and is required for the <u>synthesis of proteins</u>	
E7	rRNA associated with small and large ribosomal subunit, needed for ribosome formation	
E8	tRNA with amino acid attached – needed for translation	
Big Idea: Glycolysis [max 2]		
F1	Phosphate needed for activation of glucose	
F2	Glucose - 1 – phosphate: (only 1) - Will not move out of the cell - Keep glucose levels in the cell low - Raise energy level of glucose for substrate level phosphorylation	
Big Idea: Cell signalling [max 3]		
G1	Phosphate needed for phosphorylation of RTK, activating it	
G2	Phosphate needed for conversion of GDP to GTP for activation of GPCR	
G3	<u>Inositol triphosphate (IP₃)</u> cause <u>ligand-gated calcium channels</u> for signal transduction	
G4	Phosphorylated proteins trigger signal transduction	
G5	Phosphorylation cascade leads to signal amplification	
Big Idea: O&C Prokaryotes		
H1	Phosphate found in cAMP binds to CAP, increases transcription of Lac operon	
H2	Phosphate in ATP needed to provide energy for the transportation of <u>DNA</u> <u>moves across the plasma membrane</u> during transformation	
QWC awarded only if points come from minimum 4 big ideas.		

OR

- 5 (a) Biological specificity is one of the most widespread and characteristic properties of living organisms.

Biological specificity is most pronounced and best understood at the cellular and molecular levels of organization. Using named examples, explain the importance of shapes fitting together in cells and organisms. [13]

[Define biological specificity] **[Max 1]**

1. **[Def: Definition]** Shapes of molecules is complementary to shape of the other molecule, hence they can bind to each other.

[Proteins] **[Max 1]**

2. **[PHbH: Haemoglobin Haem group]** Oxygen binds to haem group of haemoglobin
→ Oxygen binds and dissociates → Transport of oxygen

[Enzymes] **[Max 3]**

3. **[ELK: Lock and Key]** Lock and key hypothesis → Shape of active site is exactly complementary to shape of substrate

4. **[EAS: Active site]** Shape of active site is complementary to shape of substrate → Bind to form ES complex → Catalytic function (Catabolic / Anabolic) → Product formation

5. **[EIF: Induced Fit]** Induced fit hypothesis → Shape of active site is complementary to shape of substrate, but slight conformational change to fit more snugly
[NOTE: Phrase the above statement very carefully! The substrate and enzyme are complementary in shape.]

6. **[EE: Enzyme Example]** Named example of enzyme (e.g. maltase binds and hydrolyzes maltose to α-glucose)

7. **[EP: Product]** Once product is formed → Different shape → No longer remain bound at active site

8. **[EAI: Allosteric site]** Shape of activator / inhibitor is complementary to shape of allosteric site → Change conformation of enzyme and active site → Increase / Decrease rate of enzymatic reaction

9. **[ECI: Competitive inhibitor]** Shape of competitive inhibitor is complementary to shape of active site → Compete with substrate for active site and bind at active site → Decrease rate of enzymatic reaction

10. **[ENCI: Non-competitive inhibitor]** Shape of non-competitive inhibitor is complementary to shape of allosteric site / binds at site away from active site → Changes conformation of enzyme and active site → Decrease rate of enzymatic reaction / Regulate enzyme activity and conserve energy or resources

[Transport] **[Max 1]**

11. **[TP: Transport protein]** Specific transport proteins (channel protein, carrier protein) / Shape of binding site of carrier protein is complementary to shape of molecule → Facilitated diffusion / Active transport

12. **[TRME: Receptor-mediated endocytosis]** Shape of receptor is complementary to shape of molecule → Receptor mediated endocytosis

[Cell Cycle] **[Max 1]**

13. [CSF: Spindle fibre] Attachment of spindle fibre to kinetochore on chromosome → Arrangement of chromosome in metaphase / Separation of chromosome or chromatids in anaphase
14. [CCDK: CDK] Shape of CDK is complementary to shape of cyclin → Stimulate cell division

[Photosynthesis / Respiration]

[Max 2]

15. [EP / ER: Photosynthesis / Respiration] Shape of ATP synthase is complementary to shape of ADP and P_i → Synthesize ATP → Drive metabolic reaction
16. [EP: Photosynthesis] Shape of NADP reductase is complementary to shape of $NADP^+$ and H^+ → Synthesize NADPH → Carbon reduction in Calvin cycle
[Accept: Other enzymes with substrates and functions stated]

[Nucleic acid: Complementary base pairing]

[Max 2]

17. [NDNA: DNA] Complementary base pairing between DNA strands → Formation of double-stranded DNA helix → Stability of structure of DNA molecule
18. [NRep: Replication] Complementary base pairing between DNA parental and daughter strands → Template for DNA replication
19. [NPr: Proofreading] Complementary base pairing between DNA parental and daughter strands → Template for DNA repair / proofreading
20. [NT: Telomere] Complementary base pairing between free DNA nucleotides and telomerase RNA → Template for elongation of telomeres
[Accept: Telomere (DNA) and telomeric RNA]
21. [NRNA: RNA] Complementary base pairing within tRNA → Structure of tRNA
[Accept: rRNA, telomerase RNA]
22. [NTc: Transcription] Complementary base pairing between DNA template and mRNA → Template for mRNA synthesis / transcription
23. [NPTM: Post-transcriptional modification] Complementary base pairing between snRNA of spliceosome and DNA sequence at splice site → RNA splicing
24. [NTI: Translation] Complementary base pairing between codon of mRNA and anti-codon of tRNA → Translation / Synthesis of proteins

[Nucleic acid: Shape complementary in shape]

[Max 3]

25. [PRep: Replication] Shape of DNA binding domain of helicase is complementary to shape of DNA sequence at Ori → Separate two parental strands
26. [PRep: Replication] Shape of DNA binding domain of primase / DNA polymerase / ligase is complementary to shape of 5' phosphate group of free (DNA) nucleotide and 3' OH group of adjacent nucleotide, as well as parental strand → Synthesize primers / Formation of phosphodiester bond between nucleotides of daughter strand / Seals nick between DNA fragments during replication
[Accept: Proofreading ability of DNA polymerase]
27. [PTF: Transcription Factors] Shape of DNA binding domain of general transcription factor / activator / repressor is complementary to shape of DNA sequence at promoter / enhancer / silencer → Increase / Decrease rate of transcription
28. [PTc: Transcription] Shape of DNA binding domain of RNA polymerase is complementary to shape of 5' phosphate group of free RNA nucleotide and 3' OH group of adjacent nucleotide, as well as template strand → Formation of phosphodiester bond between nucleotides of mRNA during transcription

29. [PAA: Amino acid activation] Shape of amino acid + ATP + tRNA is complementary to shape of active site of aminoacyl-tRNA synthetase → Amino acid activation
30. [PIF: Translation Initiation Factors / Repressors] Shape of eIF / translational repressor is complementary to shape of mRNA sequence at 5' UTR / 3' UTR → Increase / Decrease rate of translation
31. [PPT: Peptidyl transferase] Shape of peptidyl transferase is complementary to shape of aminoacyl-tRNA → Formation of peptide bond between amino acids to form polypeptide during translation
32. [PRF: Release factor] Shape of release factor is complementary to shape of mRNA sequence at stop codon → Termination of translation

[Bacteria]

[Max 2]

33. [PRE: Restriction enzyme] Shape of restriction site of restriction enzyme is complementary to shape of sequence at restriction site → Cut DNA at restriction site / Hydrolyse phosphodiester bonds
34. [PO: Operon] Shape of inducer / corepressor is complementary to shape of repressor → Switch on / off operon → Rapid response to changes in environment
35. [POR: Repressor] Shape of DNA binding domain of repressor is complementary to shape of DNA sequence at operator → Binding of active repressor at operator → Prevent transcription
36. [PCAP: CAP binding site] Shape of CAP is complementary to shape of CAP-binding site → Switch on operon
[Accept: cAMP and CAP are complementary in shape]

[Cell Signalling]

[Max 2]

37. [SR: Receptor] Shape of ligand is complementary to shape of receptor (e.g. insulin or glucagon receptor) → Signal reception → Cell signalling pathway
38. [SST: Signal transduction] Shape of second messenger is complementary to shape of effector protein → Cellular response
[Accept: Any relay proteins, effector proteins, Ras]

[Pathogens and antibiotics]

[Max 2]

39. [DP: Phage] Shape of tail fibre in phage is complementary to shape of receptors on surface of *E coli* → Binding of phage / Entry of phage DNA into host cell
[Reject: Entry of phage]
40. [DI: Influenza] Shape of haemagglutinin in influenza virus is complementary to shape of sialic acid receptors on respiratory epithelial cells → Endocytosis / Entry of influenza virus into host cell
OR
[DH: HIV] Shape of gp120 / gp41 in HIV is complementary to shape of CD₄ receptors on immune cells / T-helper cells → Fusion of HIV viral envelope with plasma membrane of CD₄⁺ immune cells
[Accept: Protease, Integrase, Reverse transcriptase]
41. [DB: Bacteria] Shape of antigen in pathogen (e.g. Pathogen Associated Molecular Pattern) is complementary to shape of receptors (e.g. Pattern Recognition Receptor) on immune cells → Elicit immune response / Trigger inflammatory response
[Accept: Enzymes in transformation, conjugation, transduction]
42. [DAb: Antibiotics] Shape of penicillin is complementary to shape of transpeptidase in bacteria → Inhibit formation of peptide cross-links between peptidoglycan → Kill bacteria

[Immunology]

[Max 2]

43. [IB: B-cell receptor] Shape of B cell receptor is complementary to shape of antigen → Elicit immune response / Activation of B cell
 [Accept: Epitope of antigen]

44. [IT: T-cell receptor] Shape of T cell receptor is complementary to shape of antigenic peptide on MHC of antigen presenting cell (including B cell) → Activation of T and B cell OR Proliferation and activation → Adaptive immune response OR Formation of memory T and B cells
 [Accept: NK cells]

45. [IIg: Antibodies] Shape of antigen-binding site / variable region of antibodies is complementary to shape of antigen → Opsonization / Agglutination / Neutralisation of toxins / Complement activation / Antibody-dependent cytotoxicity

46. [IV: Vaccine] Shape of antigen in vaccine is complementary to shape of receptors on immune cells → Elicit immune response

[Others]

[Max 1]

47. [CC: Cell-cell adhesion / Cell-cell recognition] Shape of glycoprotein / glycolipid / protein of one cell is complementary to the receptors of another cell → Cell-cell adhesion / Cell-cell recognition

[QWC]

[Max 1]

48. [QWC] Paragraphing + At least 1 example of protein, enzyme and nucleic acid

(b) Describe the effects of different types of mutations on the proteins of eukaryotes. [12]

1 mark EACH:

[Gene Mutation]

[Max 6]

1. [G: Gene mutation] Gene mutation is the change in nucleotide sequence / codon, and subsequently amino acid sequence
2. [S: Substitution] Substitution: Replacement of one or more nucleotides
3. [SM: Silent mutation] Silent mutation → Same amino acid → Protein structure and function not affected
4. [NC: Non-coding] Mutation in non-coding region → Same amino acid → Protein structure and function not affected
5. [SS: Splice site] Mutation in splice site → Spliceosome unable to bind → Unable to splice → Non-functional protein
6. [MM: Missense mutation] Missense mutation → Different codon that codes for different amino acid → Change in primary, secondary and tertiary structure → Protein structure and function may be affected / Non-functional protein / Solubility affected
7. [MC: Mutation at crucial site] If mutation occurs at crucial site / catalytic site / active site → Protein / Enzyme structure and function affected
 [Accept: Mutation in control elements, centromere]
8. [MN: Mutation at non-crucial site] If mutation occurs at non-crucial site → Protein / Enzyme structure and function not greatly affected
9. [NM: Non-sense mutation] Nonsense mutation → Stop codon → Premature termination of translation → Truncated / Shorter non-functional protein

10. **[ID: Insertion, Deletion]** Insertion/ Deletion: Addition / Removal of nucleotide
11. **[FSM: Frameshift mutation]** Insertion/ Deletion: Addition / Removal of nucleotide (non-multiples of 3) → May result in frameshift mutations → Affects reading of codons / reading frame downstream of mutation → Sequence of amino acids downstream of mutation being completely altered → Non-functional protein / Shorter protein
12. **[IDT: Insertion, Deletion of triplet bases]** Insertion/ Deletion of nucleotide (multiples of 3) → Addition / Removal of (one) amino acid → No effect / Non-functional protein / Shorter protein
13. **[GE: Example of gene mutation]**
 - Substitution T changes to A in template strand of beta-globin gene → Hydrophilic glutamine changes to hydrophobic valine in haemoglobin → Hydrophobic region → Polymerization of HbS / Crystallization of HbS into rod-like fibres → Sickle cell anaemia
 - ras → unable to hydrolyse ATP → constant activation of cell signaling → uncontrolled cell division
 - Junctional diversity: Joining of V to D, D to J segments → Increase diversity of antibody variable region / antigen binding site in antibody
 - Somatic hypermutation → Greater diversity / repertoire of B cell receptor / antibodies → Increased possibility of greater binding affinity of antibody to antigen

[Chromosomal Aberration]

[Max 6]

14. **[C: Chromosomal aberration]** Chromosome aberration is the change in structure or number of chromosome

[Changes in chromosomal structure]

15. **[D: Duplication]** Duplication → Set of genes repeated / Extra copy of genes → More protein products synthesized
16. **[Del: Deletion]** Deletion → Loss of a region of chromosome → Shorter chromosome missing certain genes → Proteins not synthesized / Loss-of-function
17. **[I: Inversion]** Inversion → Breaking and reattachment of chromosome in reverse orientation → Non-functional protein synthesized
18. **[T: Translocation]** Translocation → Breaking and joining of chromosome to another non-homologous chromosome
 WITH If chromosome is translocated to strong promoter → Overexpression of proteins (e.g. proto-oncogene → oncogene)
 OR
 If chromosome is translocated to a region which is transcriptionally not active / heavily methylated → Proteins not synthesised (e.g. mutated tumour suppressor genes)

[Changes in chromosomal numbers]

19. **[A] Aneuploidy** → Gain / Loss of one or more chromosomes → Can be lethal in animals
 WITH
Loss of certain genes → Proteins in deleted regions not synthesized
 OR
Extra copy of certain genes → More protein products synthesized
20. **[P] Polyploidy** → Gain / Loss of one or more SETS of chromosomes
 WITH

Loss of certain genes → Proteins in deleted regions not synthesized
 OR
Extra copy of certain genes → More protein products synthesized

21. **[CE: Example of chromosomal aberration]**

- e.g. Trisomy 21 → Down syndrome
- e.g. Extra X chromosome → XXY (Klinefelter syndrome)

[Gain-of-function / Loss-of-function]

22. **[GOF]** e.g. Chromosome is translocated to strong promoter → Overexpression of proteins

23. **[GOFE]** e.g. Ras proto-oncogene mutated to oncogene → Hyperactive Ras protein (intrinsic GTPase unable to hydrolyse GTP) → Unable to terminate signal transduction → Uncontrolled cell division

24. **[LOF]** e.g. Chromosome is translocated to a region which is transcriptionally not active / heavily methylated → Proteins not synthesized

25. **[LOFE]** e.g. mutated p53 tumour suppressor genes / DNA repair gene → Unable to detect or repair DNA damage / Initiate apoptosis → Uncontrolled cell division

[QWC]

[Max 1]

26. Paragraphing + Gene mutation + Chromosomal mutation

[Total: 25]